Traces

Let $U \subset \mathbb{R}^n$ be a bdd open set st $\partial U \in C^1$. Let $1 \leq p < \infty$.

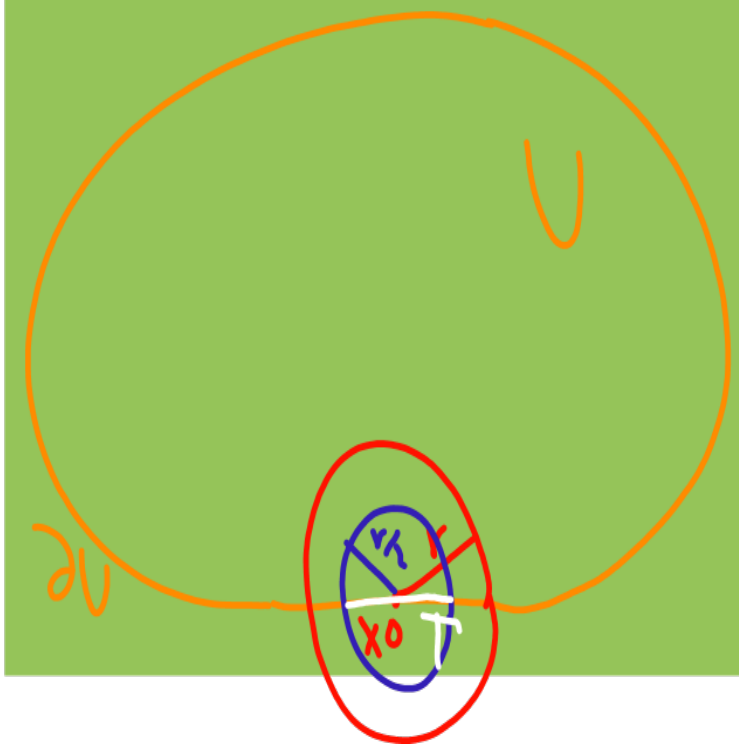
Thm: There exists a bdd linear operator (i.e., $\|Tu\|_{L^p(\partial U)} \leq C \|u\|_{W^{1,p}(U)}$ for all $u \in W^{1,p}(U)$)

$$T : W^{1,p}(U) \longrightarrow L^p(\partial U)$$

st $Tu = u|_{\partial U}$ if $u \in C(\bar{U})$.

Here Tu is called the trace of u on ∂U.

<u>Pf</u>: First suppose $u \in C^1(\bar{U})$. Let $x^0 \in \partial U$ and first consider the case where ∂U is flat near x^0, i.e., $\exists r > 0$ st



$$U \cap B_r(x^0) = \{x = (x_1, \dots, x_n) \in B_r(x^0) : x_n > 0\}$$

with a suitable coordinate system. Let $T = \partial U \cap B_{r/2}(x^0)$. Take a cut-off function

$\varphi \in C_c^\infty(B_r(x^0))$ st $0 \le \varphi \le 1$ and $\varphi \equiv 1$ on $B_{r/2}(x^0)$. Then

$$\int_{\Gamma} |u|^p \, dS \le \int_{\partial U \cap B_r(x^0)} \varphi |u|^p \, dS$$

$$= \underbrace{- \int_{\tilde{\Gamma}} -\varphi |u|^p \, dS}_{\nu^n = -1} + \underbrace{\int_{\partial B_r(x^0) \cap U} \varphi |u|^p \, dS}_{0}$$

$=0$ on $\partial B_r(x^0)$

$$\int_U u v_{x_i} \, dx = \int_{\partial U} u v \nu^i \, dS - \int_U u_{x_i} v \, dx$$

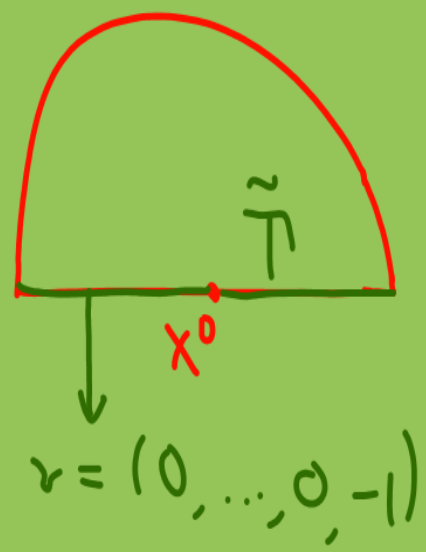
$$= -\int_{U\cap B_r(x^0)} \left(\varphi |u|^p\right)_{x_n} dx$$

$$= -\int_{U\cap B_r(x^0)} \left(\varphi_{x_n}|u|^p + p\varphi|u|^{p-2}uu_{x_n}\right) dx$$

$$\leq \int_U \underbrace{\left|\varphi_{x_n}|u|^p + p\varphi|u|^{p-2}uu_{x_n}\right|}_{\underbrace{|\varphi_{x_n}|}_{\leq C}|u|^p + p\underbrace{|\varphi|}_{\leq 1}|u|^{p-1}\underbrace{|u_{x_n}|}_{\leq |Du|}} dx$$

For $u=1$,

$$\int_U v_{x_i}\, dx = \int_{\partial U} v\nu^i\, dS.$$



$$\leq C \int_U \Big(|u|^p + \overbrace{|u|^{p-1}}^{a} \overbrace{|Du|}^{b} \Big) \, dx$$

$$\leq \frac{p-1}{p} |u|^p + \frac{1}{p} |Du|^p$$

$$\leq C \int_U \Big(|u|^p + |Du|^p \Big) \, dx$$

$$= C \, \|u\|^p_{W^{1,p}(U)}.$$

$$\|u\|^p_{W^{1,p}(U)} = \|u\|^p_{L^p(U)} + \|Du\|^p_{L^p(U)}$$

$|u|^p, \; |Du|^p$

$\tilde{p} = \frac{p}{p-1}, \quad \tilde{q} = p$

$$\frac{1}{\tilde{p}} + \frac{1}{\tilde{q}} = \frac{p-1}{p} + \frac{1}{p} = 1$$

$a, b \geq 0, \quad \frac{1}{\tilde{p}} + \frac{1}{\tilde{q}} = 1$

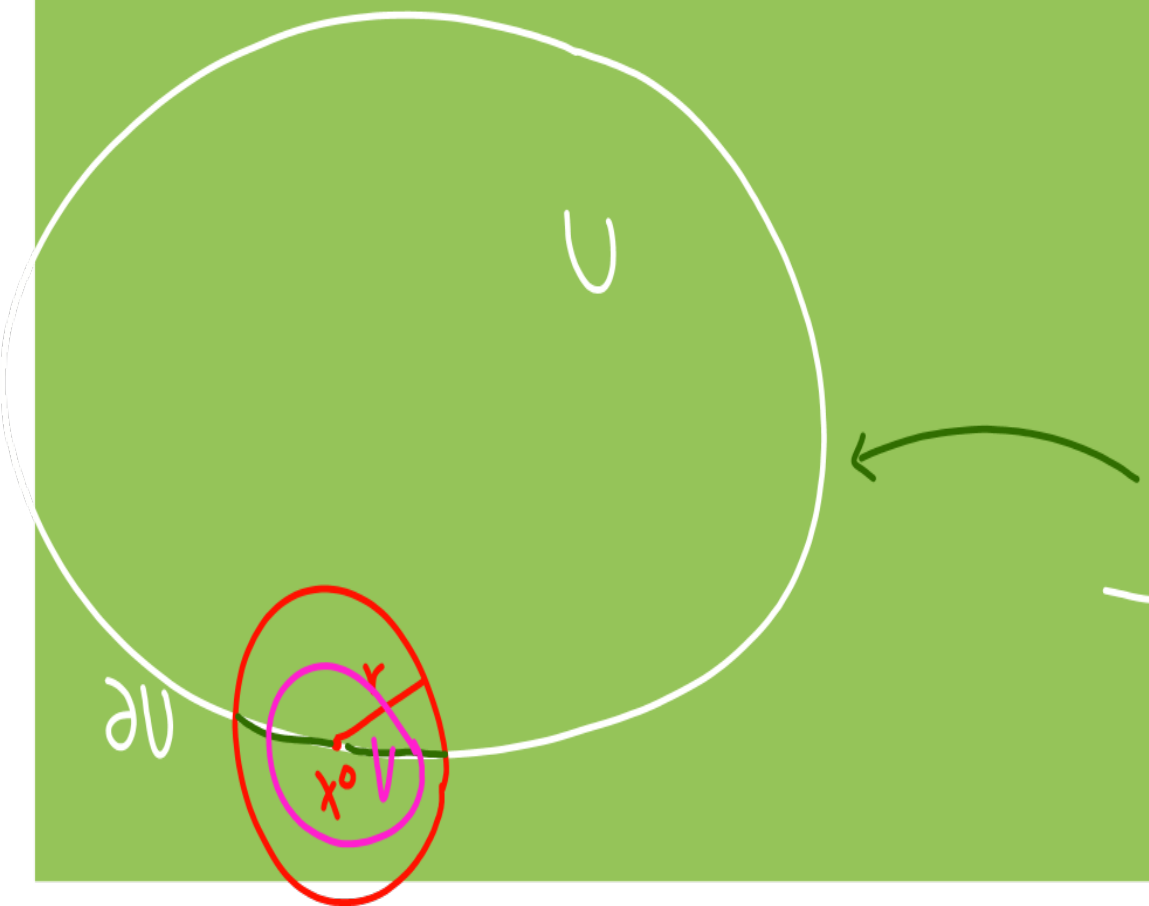
$$ab \leq \frac{1}{\tilde{p}} a^{\tilde{p}} + \frac{1}{\tilde{q}} b^{\tilde{q}}$$

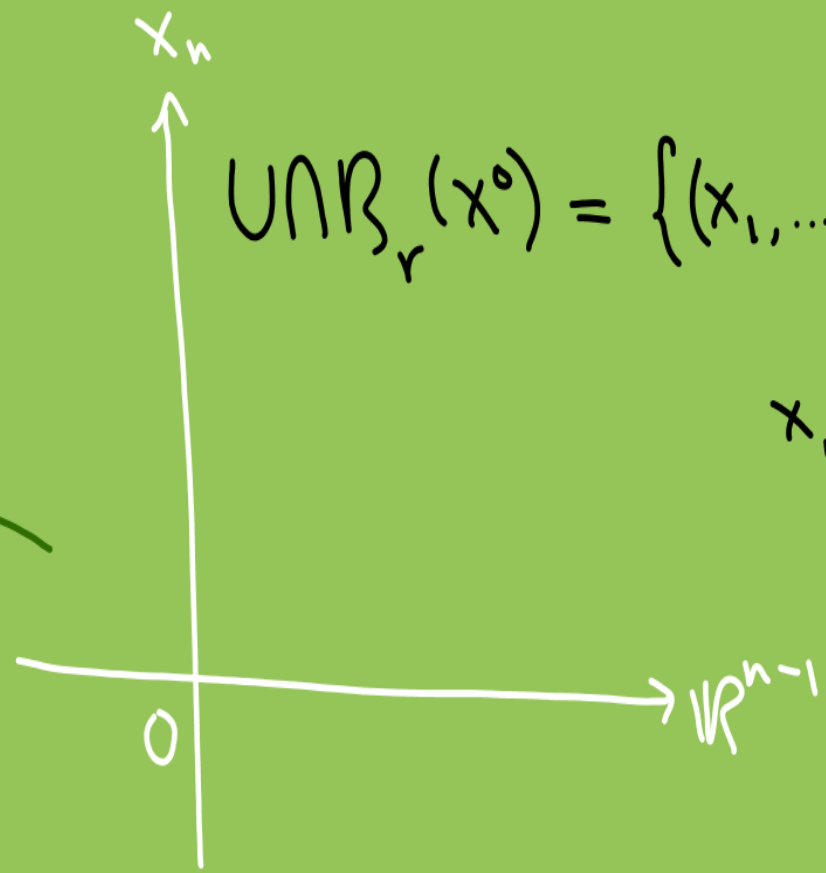
(Young's inequality)

$$\left(\varphi |u|^p\right)_{x_n} = \varphi_{x_n} |u|^p + p\varphi |u|^{p-2} u u_{x_n}$$

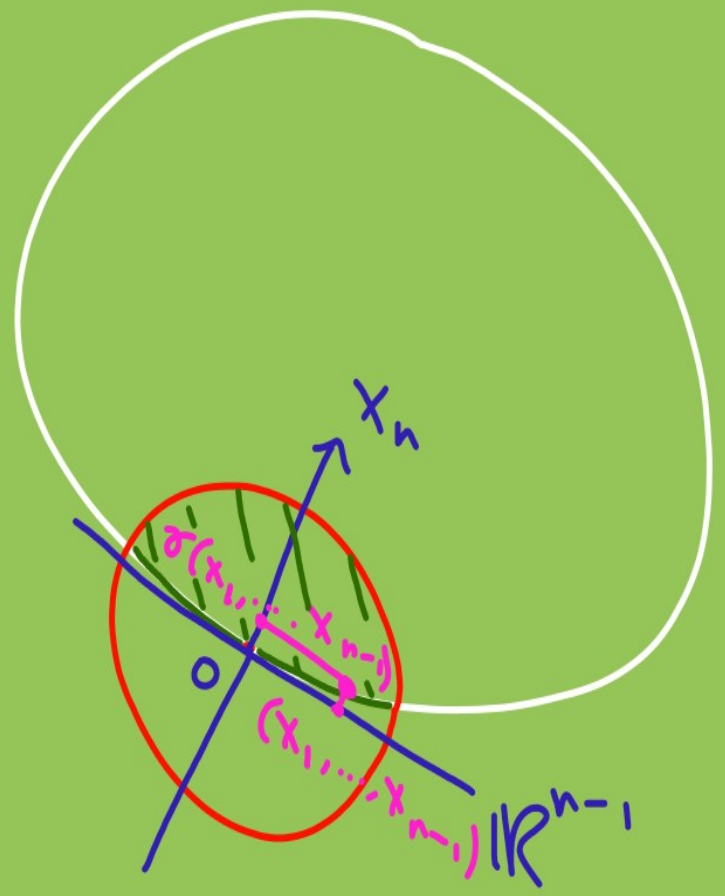
$$\left(|u|^p\right)_{x_n} = p|u|^{p-2} u u_{x_n} \begin{cases} \underline{u \geq 0:} & \left(|u|^p\right)_{x_n} = \left(u^p\right)_{x_n} = p u^{p-1} u_{x_n} \\ \underline{u < 0:} & \left(|u|^p\right)_{x_n} = \left(-u\right)^p_{x_n} = -p(-u)^{p-1} u_{x_n} \end{cases}$$

Next consider the case where ∂U is not necessarily flat near x^0. Since $\partial U \in C^1$, $\exists$ a C^1-function $\gamma : \mathbb{R}^{n-1} \longrightarrow \mathbb{R}$ st





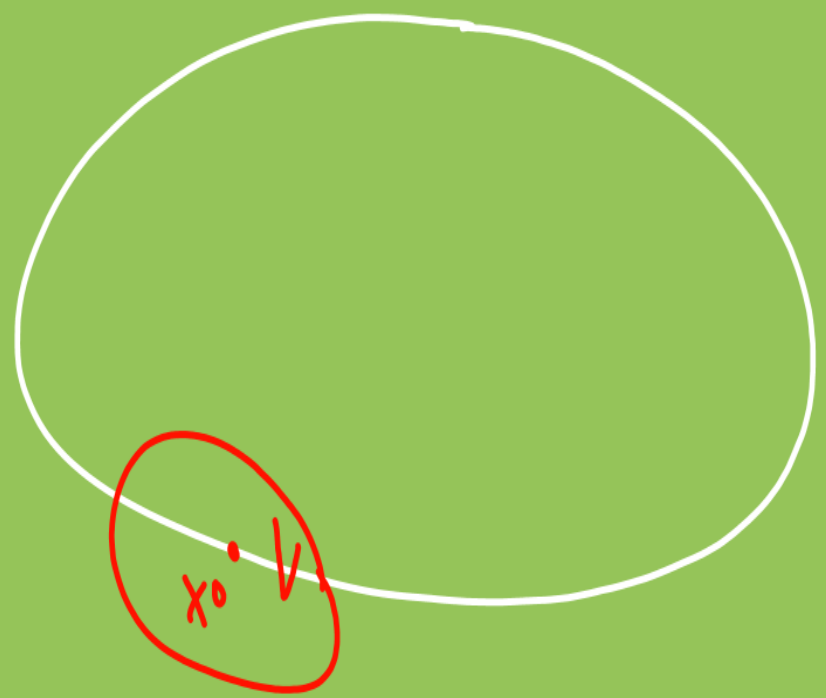
$$U \cap B_r(x^0) = \{(x_1, \dots, x_n) \in B_r(x^0) : x_n > \gamma(x_1, \dots, x_{n-1})\}.$$


x_n
$\gamma(x_1, \ldots, x_{n-1})$
0
$(x_1, \ldots, x_{n-1})$
$\mathbb{R}^{n-1}$

For some nbhd V of each $x^0 \in \partial U$,

$$\int_{\partial U \cap V} |u|^p \, dS \leq C \|u\|^p_{W^{1,p}(U)}.$$

Since ∂U is compact, $\exists$ a finite number of open sets $V_1, \ldots, V_m$

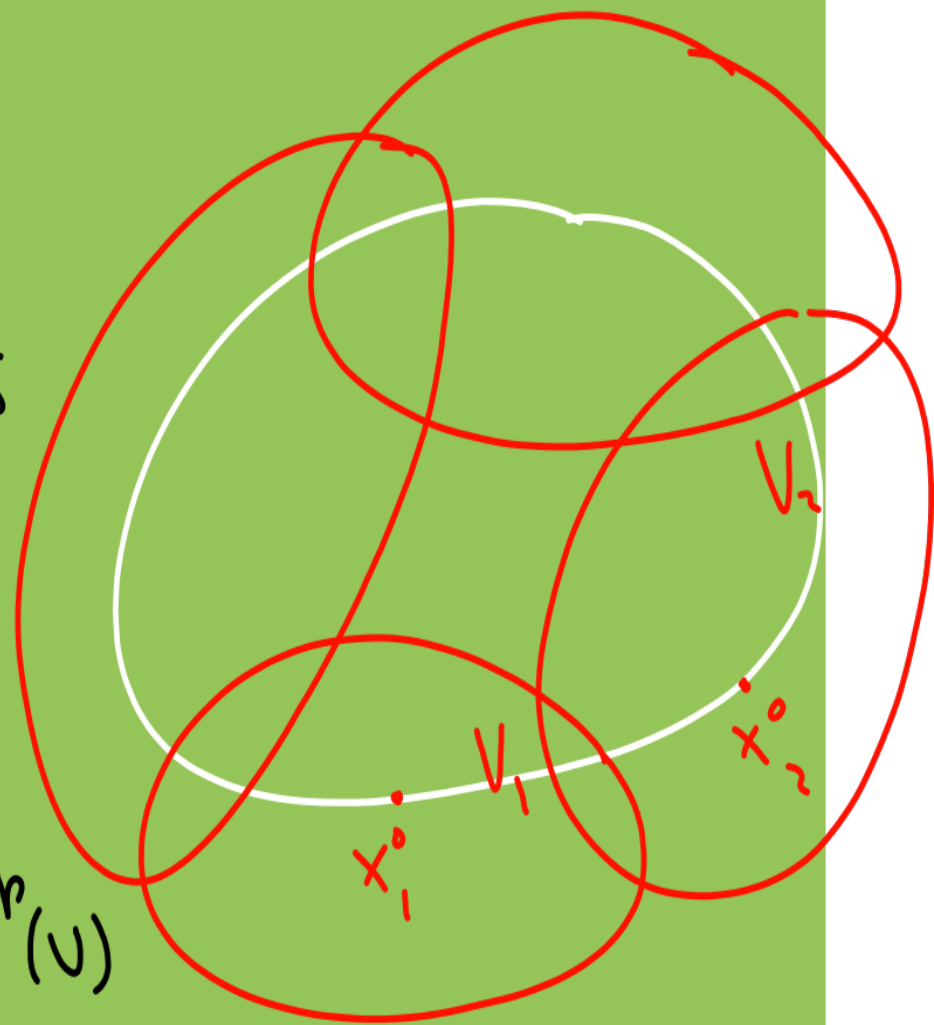


st $\partial U \subset V_1 \cup \ldots \cup V_m$ and

$$\int_{\partial U \cap V_i} |u|^p \, dS \leq C_i \|u\|^p_{W^{1,p}(U)}.$$

Then

$$\int_{\partial U} |u|^p \, dS \leq \sum_{i=1}^{m} \int_{\partial U \cap V_i} |u|^p \, dS$$

$$\leq \sum_{i=1}^{m} C_i \|u\|^p_{W^{1,p}(U)}$$

$$= C \|u\|^p_{W^{1,p}(U)}$$